A Guided Ethics Assessment for Early Prototypes in AI Education

Abstract

Students and first-time developers building AI prototypes often struggle to translate high-level ethical principles into concrete actions relevant to their specific projects. This paper introduces a Guided AI Ethics Assessment, an interactive educational tool designed to bridge this gap. Through a guided questionnaire enriched with contextual hints, real-world case studies, and analogies, the tool helps students systematically analyze the purpose, data sources, and capabilities of their project. A key objective of this process is to ensure that high-risk and critical considerations, such as those outlined in the EU AI Act, are not overlooked. Based on their responses, the system employs a "rules-as-data" architecture to deterministically generate two key educational artifacts: a prioritized Risk Assessment Report that explains the potential issues identified during the self-assessment, and a transparent Decision Trace that shows the student how their design choices led to these specific recommendations.

By transforming abstract principles into a structured, reflective dialogue, our tool makes responsible AI development a tangible, reviewable, and teachable component of AI education. We present the tool's pedagogical design, its rigorous evaluation against a diverse set of synthetic use cases, and a workflow for classroom integration, offering a practical approach to embedding ethical deliberation into the training of the next generation of AI builders.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; • **Social and professional topics** → *Codes of ethics*; • **Software and its engineering** → *Software verification and validation*; • **Applied computing** → *Interactive learning environments*.

Keywords

AI ethics, education, checklists, rules-as-data, decision tables, governance, reproducibility

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1 Introduction

As artificial intelligence systems become increasingly democratized, university students and first-time developers are no longer just consumers of AI but active builders, frequently shipping prototypes that interact with real people and their data [13]. In these fast-paced educational and project-based settings, such as a weekend hackathon or a 10-week course, ethics guidance must be delivered early, be specific to the project at hand, and be immediately actionable [1, 9]. However, students are often directed to lengthy, general-purpose documents and high-level principles, such as the EU Ethics Guidelines for Trustworthy AI [2] or the UNESCO recommendation on the Ethics of Artificial Intelligence [12], which are difficult to operationalize for their specific projects [8, 10]. This creates a critical gap between well-meaning ethical principles and the concrete, technical decisions students must make [7].

To bridge this gap, this paper introduces a **Guided AI Ethics Assessment**, an interactive educational tool designed to help students discover and reflect on the ethical dimensions of their work. Instead of requiring students to complete a static form, the tool engages students in a guided dialogue about their project. The system's "rules-as-data" architecture processes their responses against versioned decision tables. This engine is intentionally designed for high recall to serve as a safety net, ensuring that potential high-risk and critical issues are not overlooked. The process deterministically generates two key educational artifacts: (1) a prioritized **Risk Assessment Report** summarizing the potential issues identified, and (2) an auditable **Educational Decision Trace** that transparently shows how their answers led to each outcome [4, 9].

Our primary design goal is to empower students to practice ethical reflection from the very first day of a project [14]. The tool is intended to provide a structured and repeatable process for self-assessment, making a student's ethical considerations transparent and reviewable. By embedding this reflective process early in the development lifecycle, our approach helps make ethical practice a tangible and integral component of AI education [1, 3]. The primary contributions of this work are:

(1) **A "rules-as-data" backbone** with typed slots and versioned decision tables for the deterministic generation of educational ethics reports. (2) **A student-friendly interactive questionnaire** that covers foundational topics and includes targeted questions for specific AI capabilities, such as biometrics, generative media, and autonomous systems. (3) **Educational and auditable outputs**: a prioritized *Risk Assessment Report* and an *Educational Decision Trace* linking each finding to specific student inputs. (4) **A pragmatic evaluation protocol** that verifies the system's high-recall performance and robustness using simulation over synthetic profiles. (5) **A set of open materials** designed for straightforward classroom adoption.

2 Background and Design Objectives

2.1 Motivation and Context

University teams increasingly prototype AI systems that touch real people and data. High-level frameworks, such as the EU HLEG Trustworthy AI Guidelines, the OECD AI Principles, and the UNESCO Recommendation, offer widely accepted pillars (e.g., human oversight, transparency, fairness, accountability, robustness) [6, 11, 12].

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Security-oriented bodies emphasize implementable controls and risk management [3]. However, students and first-time builders often struggle to translate these principles into concrete steps under tight timelines. Decades of experience with checklists in safety-critical fields suggest that short, context-aware instructions can reliably raise the quality and consistency of practice [5]. We aim to bring that pragmatism to student AI projects without requiring expert committees or long review cycles.

2.2 Problem Setting

Our work targets the specific environment of early-stage educational builds, such as coursework, labs, and hackathons. This setting is defined by a distinct set of challenges. First, **time is highly constrained**, meaning any ethics guidance must be delivered during the initial design phase, not as an afterthought during grading or after deployment. Second, **projects are heterogeneous**, with computer vision, natural language processing, and embodied systems often developed side-by-side; a generic, one-size-fits-all checklist would either over-warn on irrelevant risks or miss critical domain-specific issues. Third, **assessment must be fair and consistent**, requiring instructors to have outputs that are auditable and reproducible rather than subjective. Finally, **uncertainty is a key feature** of early development, so the guidance mechanism must accommodate tentative decisions and allow for *unknown* inputs without forcing students to make premature or unsupported claims.

2.3 Design Objectives

To address the challenges of teaching applied AI ethics in project-based settings, three core objectives guide the design of our tool. These objectives ensure the tool is not only functional but also pedagogically effective and respectful of its users.

- **O1 - To Be Guided & Responsive:** The tool must provide an interactive educational experience, not a static form. It guides students through the self-assessment using a progress bar to manage expectations and provides contextual aids like hints and analogies to support learning. The design is also responsive to user needs; a student can terminate the assessment at any point and generate a partial report, ensuring that even time-constrained users can receive immediate feedback on the most critical, early-stage questions.
- **O2 - To Be Transparent, Private, & Repeatable:** The assessment process must be deterministic and transparent. The same set of answers must always produce the same report to ensure the experience is fair, consistent, and repeatable. Every finding must be auditable via an *Educational Decision Trace* that links recommendations directly back to the student's specific answers. This commitment to transparency extends to privacy; the tool is designed to operate entirely client-side, with no student data being stored or transmitted, thereby ensuring privacy by design.
- **O3 - To Be Comprehensive & Prioritized:** The tool must cover a comprehensive range of ethical topics, from foundational principles to specific AI capabilities like biometrics and generative AI. However, to avoid overwhelming the student, the final *Risk Assessment Report* must be focused and actionable, grouping findings by severity (e.g., Critical, High, Medium Risk) so students can immediately address the most important issues first.

2.4 A 'Rules-as-Data' Architecture

A central design choice of our system is to encode all guidance and logical pathways into versioned decision tables (stored as JSON files), treating our rules as data rather than as hard-coded logic. This architecture provides several key benefits for an educational context. It enhances **Transparency for Learning**, as the decision logic itself can be reviewed and discussed as course material. This also supports **Curriculum Adaptability**, allowing an instructor to update the ethical knowledge base to reflect new regulations or course objectives without altering the core application. Furthermore, this approach improves **Verifiable Integrity**; because the rules are declarative, they enable the systematic and automated testing detailed in our evaluation section, a practice recommended for building trustworthy AI [12]. Finally, this ensures **Reproducibility**, guaranteeing a deterministic and fair self-assessment experience for every student.

2.5 Scope and Boundaries

It is important to clarify the scope of our system. This tool is designed to scaffold ethical reflection and enhance pedagogy; it *does not* certify legal compliance or guarantee system safety. It is intended to complement, rather than replace, formal institutional processes such as an Institutional Review Board (IRB) or a Data Protection Officer (DPO) review where those are required. The use of ternary inputs (*yes/no/unknown*) helps curb false precision in early project stages but may occasionally under-trigger guidance; we expect students to re-engage with the assessment as their projects mature and technical details become clearer. While the tool's underlying knowledge base is designed to be comprehensive, its coverage will inevitably lag in niche or rapidly emerging domains. However, the open, "rules-as-data" nature of the system is designed to support continuous community-driven improvement.

3 System Overview

The architecture, illustrated in Figure 1, is designed around a dialogue between a student and a knowledge-based engine, culminating in the generation of personalized learning artifacts. The system is composed of three core components: (1) the **Student/User Interaction Loop**; (2) the central **Guided Assessment Engine** that powers the experience; and (3) the final **Generated Educational Artifacts** that serve as the output.

3.1 The Interactive Assessment Process

The process is initiated by a student engaging in a guided self-assessment. The Guided Assessment Engine presents a series of questions that are dynamically enhanced with educational guidance. This guidance is drawn from a **Pedagogical Content Library** containing contextual hints, analogies, and case studies to aid comprehension. The student reflects on their project and provides a response, which is sent back to the engine for real-time evaluation, continuing the interactive loop.

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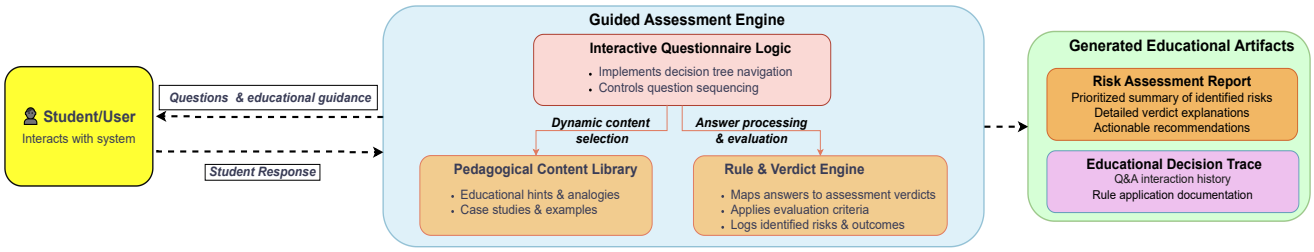

Figure 1: System Architecture: An Interactive Dialogue Model

3.2 The Guided Assessment Engine

The heart of the system is the Guided Assessment Engine, which operates on the "rules-as-data" principle. It contains three key logical components that work in concert:

- **Interactive Questionnaire Logic:** This component manages the conversation, controlling the sequence of questions based on the underlying decision tree structure and the student's previous answers.
- **Pedagogical Content Library:** This is the source of all educational content, such as hints and examples, that are dynamically selected to enrich the questions.
- **Rule & Verdict Engine:** As the student provides answers, this engine processes them in real-time. It maps responses to predefined assessment verdicts, applies evaluation criteria, and logs all identified risks and outcomes.

These components are unified by the system's logic, which is structured as a versioned decision tree. Each node in this tree represents a question and contains the question text, a set of possible answers, and rules that deterministically map a student's response to the next relevant question. For instance, if a student answers "yes" to the question about their project being in a high-risk domain, the rules in that node direct the engine to follow a specific branch of follow-up questions about the system's influence. This node-and-rule structure is what allows the assessment to be both guided and precisely tailored to each project.

This modular design ensures that the ruleset, which covers topics from EU AI Act risk classification to fairness and capability-specific risks, can be updated without changing the core application logic.

3.3 Outputs: Educational Artifacts

Once the interactive assessment is complete, the process deterministically generates two distinct educational artifacts:

(1) **The Risk Assessment Report:** This is the primary learning output for the student. It is a prioritized summary of the risks identified during the session, complete with detailed explanations and actionable recommendations for their project.
(2) **The Educational Decision Trace:** This artifact provides a transparent, auditable record of the session. It includes a full history of the question-and-answer interaction and documents how each of the student's answers was mapped to a specific rule, fulfilling our objective for transparency and serving as a powerful pedagogical tool.

4 System Implementation

This section details the technical implementation of the conceptual architecture presented in Section 3. We first describe the specific schema of the "rules-as-data" backbone—the structured JSON files that encode the system's logic. We then explain how this data structure is rendered into the user-facing interface shown in Figure 2, bridging the gap between abstract rules and the interactive student experience.

4.1 The Rules-as-Data Backbone

The engine's logic is defined by a series of nodes within a versioned JSON file, where each node represents a single question and its associated logic. The specific anatomy of a node is key to the system's dual function as both a risk assessment tool and a teaching instrument. Each node object contains:

- **Question and Answer Logic:** A `text` field for the question, an `answer` object with the user's options, and a `rules` array that maps each answer to an immediate `verdict` and the `next` node ID to continue the assessment.
- **Pedagogical Content:** A `learning_content` object containing the *significance* of the question, a real-world *case study* to illustrate the risks, and a direct *guideline* from a relevant regulation (e.g., EU AI Act, GDPR).

A key design principle is encoded directly in these rules: when a user selects "I'm not sure," the engine follows the worst-case scenario path, treating the answer as a potential risk. This "default-to-caution" approach ensures high recall for risk detection and turns uncertainty into a pedagogical moment, prompting further investigation by the student.

4.2 The User Interface

The abstract structure of the JSON backbone is rendered into the user-friendly web interface shown in Figure 2. The interface is designed to support the student's workflow and enhance the pedagogical experience. The main question and answer options from a node occupy the primary panel. The rich information from the `learning_content` object is presented in a collapsible side panel, allowing students to access simple explanations, detailed case studies, or direct links to regulations as needed, without overwhelming the main interface. In line with the principles the toolkit espouses, it also incorporates privacy-by-design; all processing occurs client-side and no personally identifiable information is stored by the system.

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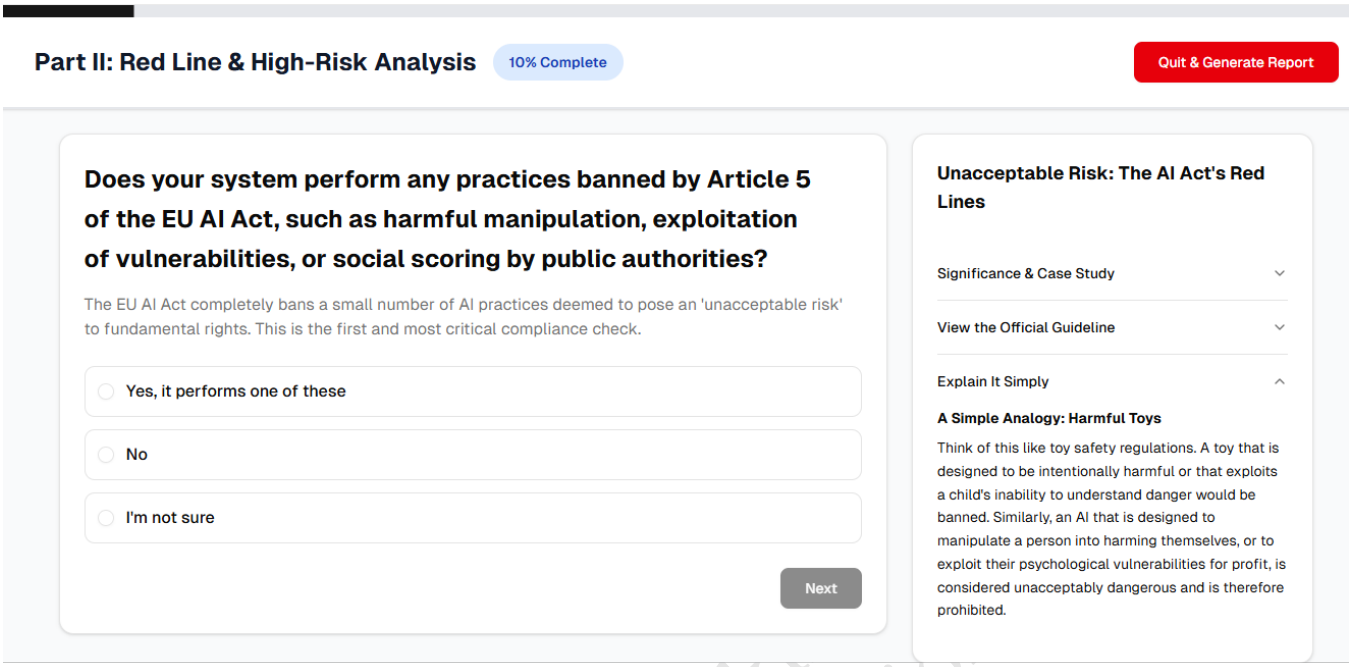

Figure 2: A screenshot of the user-facing toolkit interface. The left panel presents the current question and answer options. The right panel contains the collapsible, context-aware pedagogical content, including case studies and simple analogies, which are loaded directly from the `learning_content` object in the JSON rule file. Key UI features include the progress bar (top) and the option to generate a report at any time (top right).

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5 Evaluation and Results

To rigorously validate our AI ethics toolkit, we conducted a quantitative evaluation using a purpose-built synthetic dataset. This section details the dataset's structure, the evaluation process, the metrics used, and a multi-faceted analysis of the toolkit's performance. Our analysis examines both the overall effectiveness and a granular breakdown of its performance across different risk categories, providing a comprehensive assessment of the toolkit's capabilities.

5.1 Evaluation Dataset

The foundation of our evaluation is a dataset of 70 synthetic vignettes, each describing a unique and plausible AI system. The vignettes were intentionally designed to be diverse, ensuring the toolkit was tested against a wide spectrum of contemporary AI challenges. This section details the composition of the dataset and the methodology used to create and label it.

5.1.1 Domain and Risk Coverage. The dataset spans a variety of high-impact domains, as summarized in Table 1. This breadth ensures that our evaluation reflects the toolkit's applicability to the diverse contexts in which students and practitioners are likely to deploy AI. Furthermore, each vignette is assigned one of four distinct risk levels. This stratification, detailed in Table 2, allows for a nuanced performance analysis. The inclusion of "Critical" risk vignettes, which depict clear violations of ethical principles, is particularly important for testing the toolkit's core safety function.

5.1.2 Vignette Creation and Ground Truth Labeling. The vignettes were developed by synthesizing real-world AI applications, prominent ethical case studies found in academic literature, and the specific high-risk categories defined in regulations such as the EU AI Act. This approach ensures the dataset is grounded in authentic, contemporary challenges.

The ground truth for each vignette—the set of relevant ethical sections—was established through a rigorous labeling process conducted by the authors. For each case, the vignette's description and its structured metadata were analyzed to map it to the canonical ethical categories used by the toolkit (e.g., *Fairness & Bias, Safety & Robustness*). This mapping was guided by the principles outlined in foundational documents like the EU's Ethics Guidelines for Trustworthy AI [2] and the UNESCO Recommendation [12], ensuring that the ground truth labels are consistent and well-founded.

5.2 Evaluation Methodology and Metrics

The evaluation was conducted by simulating a student using the toolkit for each of the 70 vignettes. To ensure scientific validity and reproducibility, a deterministic script translated each vignette's structured metadata into a standard set of answers for the toolkit's questions. This automated approach removes subjective variability and allows for a precise assessment of the toolkit's underlying logic.

We approached the evaluation as a multi-label classification task, measuring performance with **Precision** (accuracy of flagged issues),

Table 1: Distribution of Vignettes by General Domain

General Domain Category	Vignette Count	% of Total
Tech, Media, & Consumer (e.g., Recommenders, IoT, Generative AI)	31	44.3%
Healthcare & Life Sciences (e.g., Diagnostics, Medical Devices)	11	15.7%
Finance, Labor, & Insurance (e.g., Lending, HR, Fintech)	11	15.7%
Government & Public Sector (e.g., Smart Cities, Justice, Social Services)	10	14.3%
Education & Academia (e.g., Proctoring, Admissions, Research Tools)	7	10.0%
Total	**70**	**100%**

Table 2: Distribution of Vignettes by Risk Category

Risk Category	Count	%
Critical	6	8.6%
High	22	31.4%
Medium	28	40.0%
Low	14	20.0%
Total	**70**	**100%**

Recall (completeness of detection), and the **F1-Score** (their balanced mean). To provide a clear picture of the results, we analyzed the scores from two standard perspectives:

- **Macro-Averaging:** The metric is calculated for each vignette individually, and then the results are averaged. This method gives every *project vignette* equal weight and is our primary reported score as it reflects performance on a typical, complete use case.
- **Micro-Averaging:** The total counts of true positives, false positives, and false negatives are summed up across all vignettes to compute a single, aggregate score. This method gives every individual *ethical label decision* equal weight.

The aggregate counts for the micro-average calculation are shown in the confusion matrix in Figure 3.

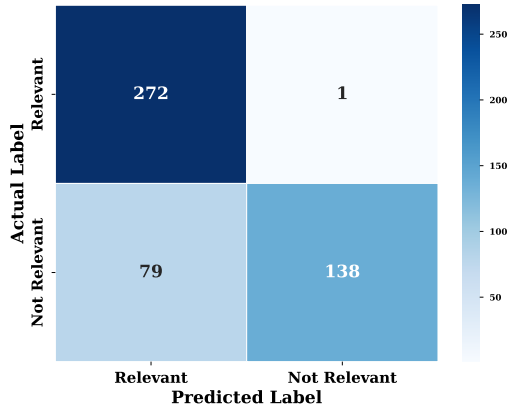

Figure 3: Aggregate Confusion Matrix Across All 490 Label Decisions

5.3 Overall Performance Analysis

The toolkit demonstrated strong overall performance, achieving a **macro-averaged F1-Score of 0.898**. The results in Table 3 resolve the two averaging methods, showing both the primary macro-averaged scores and the micro-averaged F1-score calculated from the confusion matrix.

Table 3: Overall Toolkit Performance Scores (N=70 Vignettes)

Metric Type	Score
Macro-Averaged Precision	0.871
Macro-Averaged Recall	0.941
Macro-Averaged F1-Score	**0.898**
Micro-Averaged F1-Score	0.872

The high macro-averaged recall (0.941) indicates that the toolkit is exceptionally effective at its primary goal: identifying and flagging potential ethical and compliance issues. The single False Negative across all 490 decisions underscores this strength. The toolkit is engineered to be cautious, prioritizing comprehensive risk detection—a vital characteristic for any tool intended to ensure safety and responsible innovation.

5.4 Performance by Risk Category

To gain deeper insight, we analyzed the toolkit's performance across the four risk categories. The results, presented in Figure 4, highlight how the toolkit adapts to different levels of ethical complexity.

This analysis yields several crucial findings. First, the toolkit's performance is strongest on **High-Risk** vignettes (F1-Score of 0.957), where the ethical requirements are most clearly defined by regulations and best practices. Second, for **Critical-Risk** vignettes, the toolkit achieves a perfect Recall of 1.000, successfully identifying every single unethical practice it was tested against. The lower precision in this category is an acceptable and even desirable outcome, as the toolkit correctly raises multiple, valid alarms for these inherently dangerous systems.

Finally, the analysis reveals a clear area for improvement. The toolkit's performance is lowest on **Low-Risk** projects (F1-Score of 0.730). This is not due to missing risks (Recall is still high at 0.929), but due to lower precision (0.580), where the toolkit's default "safety net" rules can generate false positives for scenarios with minimal ethical implications. This suggests that future work could focus

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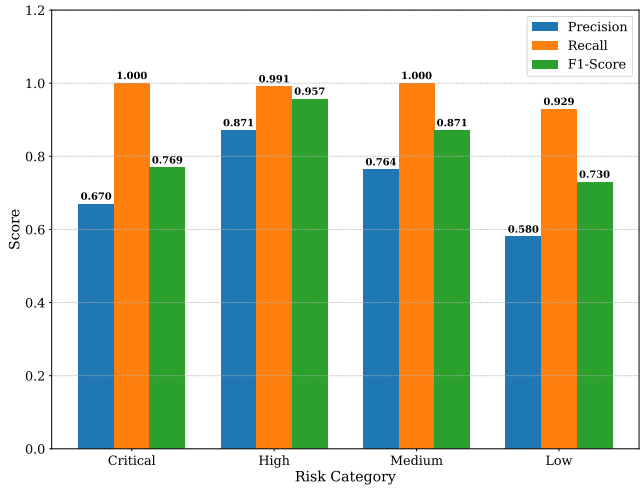

Figure 4: Toolkit Performance Metrics (Macro-Averaged) by Vignette Risk Category. Recall is notably high across all categories, especially for Critical and High-Risk cases.

on refining the logic for these low-stakes cases to reduce unnecessary warnings without compromising the toolkit's outstanding performance on high-stakes applications.

6 Classroom Integration and Discussion

The Guided AI Ethics Assessment is designed for straightforward integration into project-based computer science curricula. A typical workflow involves students using the tool in the first week of a project to generate an initial *Risk Assessment Report*. This report serves as a foundational "living document" for the project's ethical considerations. Students can then revisit the assessment at key milestones—for instance, after finalizing their dataset or before deploying a prototype—to see how their decisions have altered the project's ethical landscape.

This approach is designed to offer significant pedagogical benefits. By integrating the teaching of ethics into the applied software development lifecycle, the tool encourages students to see ethical considerations not as a bureaucratic constraint, but as a core component of building high-quality, robust systems. The immediate, context-specific feedback fosters a more nuanced understanding of responsible AI development. The auditability of the *Educational Decision Trace* further enables instructors to engage in in-depth, evidence-based conversations with students about their technical choices.

Beyond a single project assessment, the tool's artifacts can be used for other classroom activities. Instructors can have students conduct peer reviews of each other's reports and traces, fostering collaborative learning and discussion. The tool can also be used for comparative analysis, where students assess multiple design alternatives for the same problem (e.g., one using biometrics vs. one using tokens) to better understand the practical impact and trade-offs of their architectural decisions.

7 Limitations and Future Work

Our tool does not prove legal compliance or guarantee safety. While its knowledge base is comprehensive, it will inevitably lag in niche or rapidly emerging domains. Our "default-to-caution" design, which treats "unknown" answers as potential risks, intentionally prioritizes high recall but can lead to over-warning. The tool is thus a starting point for discussion, and students should be encouraged to re-engage with the assessment as their project details become clearer.

Future work will leverage the "rules-as-data" architecture to create knowledge bases for different ethical frameworks (e.g., the US NIST AI Risk Management Framework) and specialized domains like environmental impact. Additionally, having validated the tool's logic, a formal, longitudinal study is needed to assess its impact on student learning outcomes in a classroom setting. We will release all materials to support scrutiny, replication, and adaptation of this work.

8 Conclusion

This paper introduced the Guided AI Ethics Assessment, an interactive educational tool designed to bridge the critical gap between high-level ethical principles and the concrete decisions students make when building AI systems. Our approach transforms this challenge from a static compliance task into a guided, reflective dialogue. Through its transparent and deterministic "rules-as-data" architecture, the tool provides a repeatable process for self-assessment. Our evaluation, conducted on a diverse set of 70 synthetic vignettes, demonstrates the tool's effectiveness, showing a high-recall performance profile that serves as a reliable safety net for ensuring students do not overlook critical ethical considerations. Ultimately, this work provides a practical and extensible method for embedding ethical deliberation directly into computer science education, helping to foster a more responsible and conscientious generation of AI builders.

Acknowledgments

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